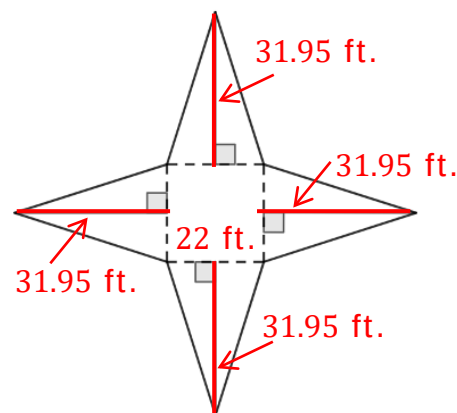


In the area that is now known as the Sudan, there is a collection of pyramids known as the Nubian Pyramids. The largest pyramid has a square base of 22 feet (ft) and stands 30 ft high. The slant height is 31.95 ft. The net of the pyramid is shown.

- Label the dimensions of the net. Include units.
- Complete the table to determine the area of each face of the pyramid.

Side	Formula	Area
Rectangular Base	$A = l \cdot w$	$A = 22 \times 22 = 484 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$
Triangular Face	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 22 \cdot 31.95 = 351.45 \text{ ft.}^2$



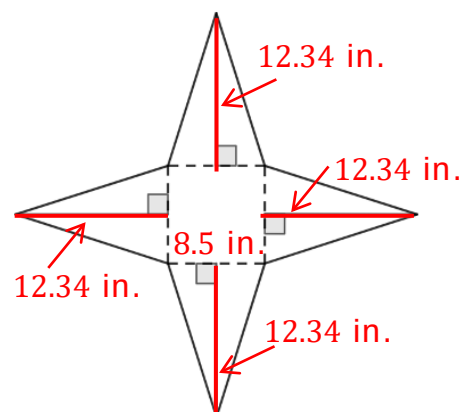
- Determine the surface area of the pyramid. Include units.

$$484 + 351.45 + 351.45 + 351.45 + 351.45 = 1,889.8 \text{ ft.}^2$$

Walt is doing a presentation on the Nubian Pyramids and wants to create a model. The base is square with a side length of 8.5 inches (in.) and is 11.59 in. high. The slant height of each triangular face is 12.34 in. The net of Walt's model is shown.

- Label the net with the dimensions of each face. Include units.
- Complete the table with the area of each face.

Side	Area (in ²)
Rectangular Base	$8.5 \times 8.5 = 72.75 \text{ in.}^2$
Triangular Face	$\frac{1}{2} \cdot 8.5 \times 12.34 = 52.445 \text{ in.}^2$
Triangular Face	$\frac{1}{2} \cdot 8.5 \times 12.34 = 52.445 \text{ in.}^2$
Triangular Face	$\frac{1}{2} \cdot 8.5 \times 12.34 = 52.445 \text{ in.}^2$
Triangular Face	$\frac{1}{2} \cdot 8.5 \times 12.34 = 52.445 \text{ in.}^2$



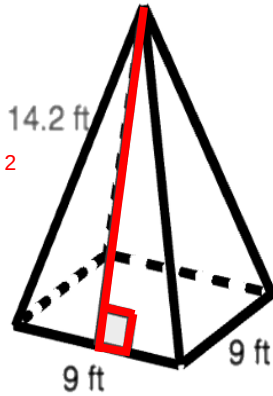
6. What is the surface area of Walt's model? Include units.

$$72.75 + 52.445 + 52.445 + 52.445 + 52.445 = 282.53 \text{ in}^2$$

For questions 7-8, find the surface area each pyramid. Include units.

7.

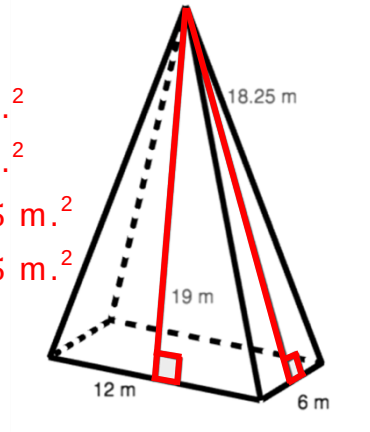
$$\begin{aligned} 1. & 9 \cdot 9 = 81 \text{ ft.}^2 \\ 2. & \frac{1}{2} \cdot 9 \cdot 14.2 \\ & = 63.9 \text{ ft.}^2 \end{aligned}$$



$$81 + 63.9 + 63.9 + 63.9 + 63.9 = 336.6 \text{ ft.}^2$$

8.

$$\begin{aligned} 1. & 12 \times 6 = 72 \text{ m.}^2 \\ 2. & \frac{1}{2} \times 19 \times 12 = 114 \text{ m.}^2 \\ 3. & \frac{1}{2} \times 19 \times 12 = 114 \text{ m.}^2 \\ 4. & \frac{1}{2} \times 6 \times 18.25 = 54.75 \text{ m.}^2 \\ 5. & \frac{1}{2} \times 6 \times 18.25 = 54.75 \text{ m.}^2 \end{aligned}$$



$$72 + 114 + 114 + 54.75 + 54.75 = 409.5 \text{ m.}^2$$

9. Calculate the surface area of a rectangular pyramid that has a square base with a side length of 11.5 inches (in.), which is the base of the triangle with a slant height of 15.13 in.

$$\begin{aligned} 1. & 11.5 \times 11.5 = 132.25 \text{ in.}^2 \\ 2. & \frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2 \\ 3. & \frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2 \\ 4. & \frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2 \\ 5. & \frac{1}{2} \times 11.5 \times 15.13 = 86.9975 \text{ in.}^2 \end{aligned}$$

$$132.25 + 4(86.9975) = 480.24 \text{ in.}^2$$

10. Calculate the surface area of a rectangular pyramid that has a rectangular base with a side length of 4 feet (ft), which is the base of a triangle with a slant height of 3.9 ft, and a side length of 5 ft, which is the base of a triangle with a slant height of 4.1 ft.

$$\begin{aligned} 1. & 4 \times 5 = 20 \text{ ft.}^2 \\ 2. & \frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2 \\ 3. & \frac{1}{2} \times 4 \times 3.9 = 7.8 \text{ ft.}^2 \\ 4. & \frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2 \\ 5. & \frac{1}{2} \times 5 \times 4.1 = 10.25 \text{ ft.}^2 \end{aligned}$$

$$20 + 7.8 + 7.8 + 10.25 + 10.25 = 56.1 \text{ ft.}^2$$